ASRC Fuel Tank Robot Daily Report

Coordinated winch control and winch accuracy, 10 November 2026

Yesterday showed that U4 plus M2 with a 1.8 mm NiTi rod in each M module can hold the 500 g head at 20 degrees while sweeping yaw from 0 to 175 degrees, using 5 degree sample poses. Two questions were left: does real rope-length (winch) control move the head smoothly between those poses, and how accurate must the winches be?

Summary of today's work

- **Coordinated winch control.** All 12 winches followed the rope lengths of the 5 degree poses together, in 20 sub-steps per segment. The head moved smoothly from 0 to 175 degrees (40 GPa rod) and 0 to 180 degrees (75 GPa rod), staying at 19.1 to 20.1 degrees elevation. All 701 and 721 states were stable, no rope went slack, and the backward sweep retraced the forward one exactly.
- **Tank clearance between poses.** The in-between states clear the full 3D tank by at least 12.6 mm with the clamp 30 mm past the opening.
- **Winch requirements.** The largest winch travel over the sweep is 69 mm and the fastest rate 1.67 mm per degree of yaw. Anchor-4 ropes barely move. Command tables are saved as CSV.
- **Winch accuracy.** One winch off by 1 mm moves the head 1.3 to 3.2 degrees. Holding plus or minus 1 degree without feedback needs about plus or minus 0.16 to 0.51 mm on every winch. The four tip ropes dominate; rope stiffness error of 10% matters little (at most 0.42 degrees).
- **Margin against slack.** Every perturbed state stayed stable, but a rope at 1 N pretension stretches only about 0.15 mm, so small winch errors can leave it slack.

Check (U4 plus M2, rod, 20 deg, head held at 20 deg)	Result	Status
Motion between the 5 deg poses, winch control	701 / 721 states, all stable, no slack, reversible	Pass
Head elevation during the sweep	19.15 to 20.10 deg	Pass
Largest head movement per sub-step (0.25 deg yaw)	1.68 mm (40 GPa), 2.53 mm (75 GPa)	Pass, no snaps
3D tank clearance between poses, clamp 0 / 30 mm	22.4 / 12.6 mm minimum	Pass
Winch accuracy for plus or minus 1 deg (99%)	0.16 to 0.51 mm	Needs feedback
Rope stiffness error plus or minus 10%	at most 0.42 deg	Minor
Ropes slack under winch error	58 of 168 single 0.5 mm cases; stability kept	Small margin

Decisions from today
Length-controlled winches can drive the full held-elevation sweep smoothly. Add head-angle feedback (for example an IMU in the head), at least for the four tip winches, and raise the rope pretension above 1 N to keep ropes taut under winch error. Anchor-4 ropes can be held fixed during steering.

Coordinated winch control between poses

For every stored 8-cell minimum-force pose (anchors [4,5,6], 1 N pretension), the length of each of the 12 winch ropes was computed with the fixed-winch rope model (elastic PA6 line, EA 10.4 kN, 1 m free length). All 12 lengths were then moved together, linearly from one pose to the next, in 20 sub-steps per 5 degree segment. Each equilibrium was solved from the previous state, giving the quasi-static motion a length-controlled drive would produce. The sweep was run forward and backward.

Check	Rod E 40 GPa	Rod E 75 GPa
States solved (forward)	701, none failed	721, none failed
Yaw range	0 to 175 deg, monotonic	0 to 180 deg, monotonic
Yaw vs commanded (linear between poses)	within 0.78 deg	within 0.90 deg
World head elevation	19.15 to 20.08 deg	19.82 to 20.10 deg
Head movement per sub-step	median 0.62, max 1.68 mm	median 0.62, max 2.53 mm
Rope tension	0.71 to 14.9 N	0.90 to 14.7 N
Stability (minimum fixed-winch eigenvalue)	all stable (0.148)	all stable (0.260)
Backward vs forward sweep	head within 0.00003 mm	head within 0.00004 mm
3D tank, clamp s = 0 / 30 mm	22.4 / 12.6 mm, all clear	23.2 / 13.4 mm, all clear
Strain / material twist	at most 0.06% / 55 deg	at most 0.07% / 50 deg

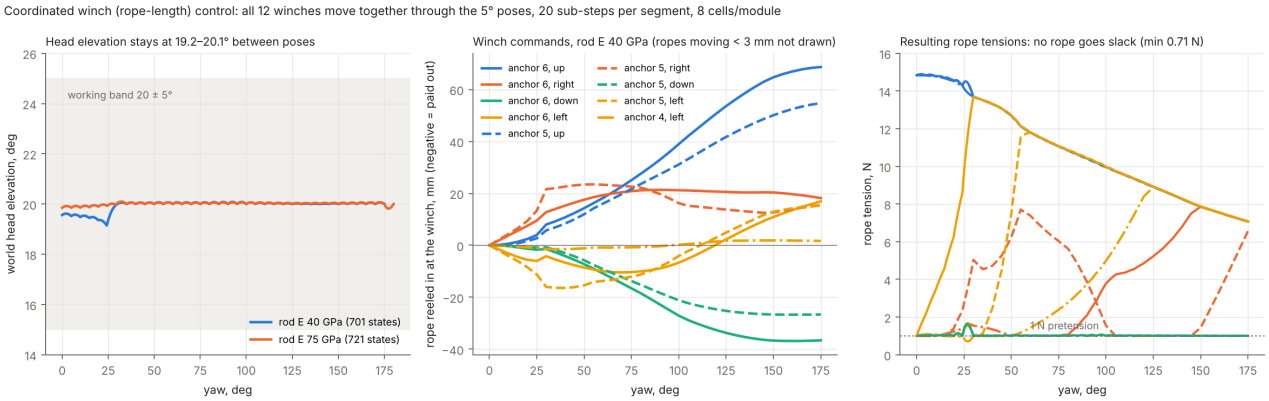


Figure 1 Left: head elevation during the continuous sweep. Middle: winch commands (rope reeled in relative to the straight pose). Right: resulting rope tensions; no rope goes slack.

- Under coordinated rope-length control the head moves smoothly and reversibly through the whole sweep at the working elevation. This closes the open question from 9 November about the motion between the sampled poses, for the quasi-static model.
- Winch requirements (40 GPa rod): largest travel 68.6 mm (tip up rope), fastest 1.67 mm per degree of yaw. The anchor-4 ropes move under 3.5 mm over the whole sweep and could be held at fixed length while steering.

Winch position error and rope stretch sensitivity

At seven poses of the coordinated sweep, each of the 12 winch lengths was offset by plus or minus 0.5 and 1 mm, one at a time, and the full nonlinear equilibrium re-solved (336 solves per rod). The resulting sensitivity matrix predicted 2,000 random cases with all 12 winches wrong at once (independent, uniform), and the worst case; the 10 worst predicted 1 mm cases per pose were re-solved with the full model. Rope EA was also changed by plus or minus 10% with the winch commands unchanged.

Rod E 40 GPa	yaw 0	60	120	150	175
One winch off by 1 mm: head angle error, deg	1.7	1.8	1.4	2.3	3.2
All 12 random plus or minus 1 mm (99%): yaw / elev, deg	1.9 / 1.9	2.3 / 2.5	2.6 / 3.2	4.3 / 3.3	6.3 / 2.1
Same: head position, mm	6	7	9	15	20
Winch accuracy for plus or minus 1 deg (99%), mm	0.51	0.41	0.32	0.23	0.16
Winch accuracy for plus or minus 2 deg (99%), mm	1.03	0.81	0.63	0.47	0.32
Winch accuracy for plus or minus 1 deg (worst case), mm	0.41	0.22	0.17	0.13	0.10

The 75 GPa rod is similar or less sensitive: plus or minus 1 degree needs 0.26 to 0.53 mm (99%).

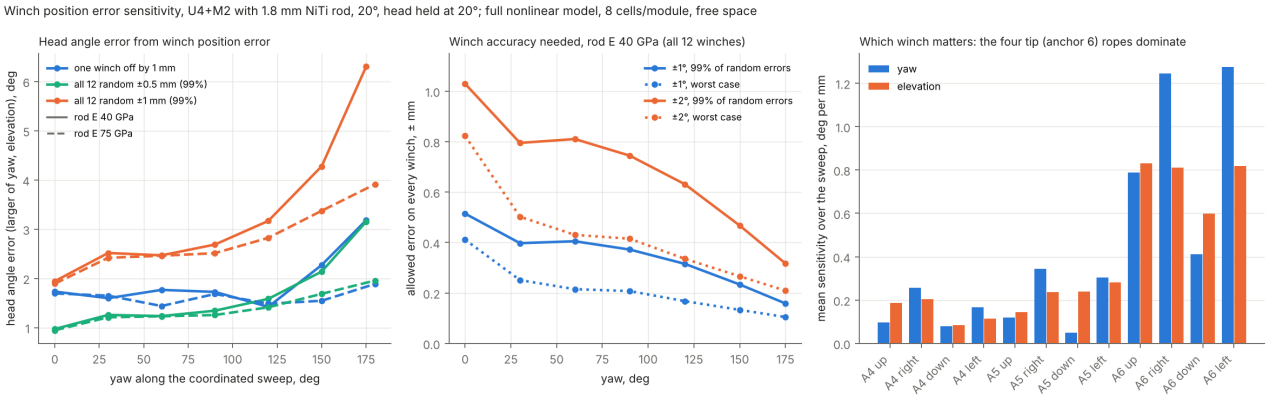


Figure 2 Left: head angle error against yaw for single and combined winch errors. Middle: winch accuracy needed for plus or minus 1 and 2 degrees. Right: mean sensitivity of each winch; the four tip (anchor 6) ropes dominate.

- Open-loop length control needs about 0.2 to 0.5 mm accuracy on every winch for plus or minus 1 degree of head angle, tightening to 0.16 mm near 175 degrees where the front curls up.
- The four tip ropes (anchor 6) give 0.4 to 1.3 degrees per mm; anchor-4 and anchor-5 ropes 0.05 to 0.35 degrees per mm. Precision matters mainly on the tip winches.
- All 672 single-winch cases and 140 combined checks stayed stable and within geometry limits. A rope at 1 N stretches only about 0.15 mm, so 58 of 168 single 0.5 mm errors left a rope slack.

Limitations

- Quasi-static model: no winch dynamics, inertia or control loop. Rope friction and stick-slip in the guides, rope creep and temperature are not modelled.
- Rope stiffness assumes 1 m of free rope from guide to winch; a different length rescales the pull-in values and the slack margin.
- Rod properties are calculated, not measured; NiTi hysteresis is not modelled. The head is a 500 g point mass with a 63 mm envelope.
- Linear prediction of combined winch errors agrees with the full model within about 1.2 degrees at 1 mm; slack ropes cause the difference.

Next steps

Priority	Task	Purpose
1	Build and test the backbone M module	Measure EA, EI, rod buckling at the guide spacing and joint slip; all results depend on these
2	Higher pretension (3 to 5 N)	Re-run the minimum-force sweep and winch sensitivity; trade peak force for margin against slack
3	Head-angle feedback concept	Sensor choice (IMU or encoder), and a closed-loop correction using the sensitivity matrix from today
4	Rope friction model	Add guide friction to the winch model and check smoothness and accuracy
5	Tank-aware steering, entry, storage, rear	As in the 9 November plan

Sources and reproduction

Scripts and data: ASRC_kevin/telescopic_planar_study/backbone_20261108 (README sections 8 and 9: coordinated_winch.py, coordinated_winch_tank.py, coordinated_winch_table.py, winch_sensitivity.py). Winch command tables: results/coordwinch_E40_winch_table.csv and coordwinch_E75_winch_table.csv. This report is generated by ASRC_kevin/telescopic_planar_study/reports/build_report_2026_11_10.py.